
LIVENEWSBENCH: Evaluating Web Search Agents with Freshly Curated News

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Abstract

Large Language Models (LLMs) with agentic web search capabilities show strong potential for tasks requiring real-time information access and complex fact retrieval, yet evaluating such systems remains challenging. We introduce LIVENEWSBENCH, a rigorous and regularly updated benchmark designed to assess the agentic web search abilities of LLMs. LIVENEWSBENCH automatically generates fresh question-answer pairs from recent news articles, ensuring that questions require information beyond an LLM’s training data and enabling clear separation between internal knowledge and search capability. The benchmark features intentionally difficult questions requiring multi-hop search queries, page visits, and reasoning, making it well-suited for evaluating agentic search behavior. Our automated data curation and question generation pipeline enables frequent benchmark updates and supports construction of a large-scale training dataset for agentic web search models, addressing the scarcity of such data in the research community. To ensure reliable evaluation, we include a subset of human-verified samples in the test set. We evaluate a broad range of systems using LIVENEWSBENCH, including commercial and open-weight LLMs as well as LLM-based web search APIs. The leaderboard, datasets, and code are publicly available at livenewsbench.com.

1. Introduction

Large Language Models (LLMs) equipped with agentic web search capabilities (OpenAI, 2025b; DeepSeek-AI, 2025b; Anthropic, 2025; xAI, 2025; Kimi Team, 2025; GLM-4.5 Team, 2025; Jin et al., 2025; Li et al., 2025) have signifi-

cantly improved performance on tasks requiring access to up-to-date or rare information. These systems can perform multi-hop web searches and online browsing to supplement their internal knowledge, enabling them to effectively handle tasks that require time-sensitive information or retrieval of obscure facts. However, despite growing interest and adoption, rigorously evaluating the search capabilities of these models remains an open challenge.

A key difficulty is disentangling the contribution of external search from a model’s internal world knowledge. Since state-of-the-art LLMs are pretrained on vast text corpora, they already encode massive amounts of world knowledge (Brown et al., 2020; Hoffmann et al., 2022). When benchmarks use static questions or question-answer pairs, it is unclear whether a model answers correctly due to successful retrieval of online information or simply by recalling memorized facts. This ambiguity limits our ability to measure true improvements from web search capabilities.

Recent work evaluating search-enabled LLMs spans three broad benchmark families, each capturing useful but incomplete signals about agentic web search. First, academic reasoning benchmarks such as Humanity’s Last Exam (Humanity’s Last Exam Team, 2025) are frequently used as a proxy for agentic search capability (OpenAI, 2025b; xAI, 2025; DeepSeek-AI, 2025b), yet they primarily measure problem-solving and domain knowledge rather than search capabilities.

Second, factual question-answering benchmarks are frequently used to evaluate search-enabled models, but many widely used datasets such as SimpleQA (Wei et al., 2024), BrowseComp (Wei et al., 2025), DeepSearchQA (Gupta et al., 2026), and TriviaQA (Joshi et al., 2017) consist of static question-answer pairs that conflate memorization with search capability. Although time-sensitive or dynamic factual QA benchmarks such as FreshQA (Vu et al., 2024) and SealQA (Pham et al., 2025) aim to address this issue, the questions within these benchmarks do not change over time, although the answers may. As a result, strong LLMs can often answer them offline through reasoning or partial memorization, limiting their effectiveness for evaluating genuine search behavior.

Third, Deep Research benchmarks (Du et al., 2025; Zhang et al., 2025; Wang et al., 2025) typically evaluate long-

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form reports using subjective criteria (e.g., completeness, insightfulness, and readability), which is complementary but does not directly target the setting where concise, verifiable factual answers are required.

These gaps motivate LIVENEWSBENCH: a contamination-limited, regularly updated benchmark designed to more cleanly separate gains due to external retrieval from those due to memorized knowledge. Our main contributions are as follows:

Automated pipeline for large-scale Q&A dataset creation and validation. In contrast to prior benchmarks that rely heavily on manual annotation, LIVENEWSBENCH is built using an automated pipeline that continuously collects recent news articles and generates question-answer pairs with minimal human intervention. This design enables frequent benchmark updates and supports the construction of a large-scale dataset. As a result, LIVENEWSBENCH can also provide a much-needed open-source training set suitable for developing agentic search models with Reinforcement Learning with Verifiable Rewards (RLVR). To ensure evaluation reliability, we additionally include a *human-verified subset* of the test set through manual validation.

Regularly refreshed and memorization-limited benchmark. Unlike prior web search benchmarks that rely on static or widely known information, LIVENEWSBENCH constructs question-answer pairs from recent news events that occur after the models’ training data cutoffs. This design substantially reduces memorization and ensures that correctly answering questions requires accessing information beyond an LLM’s internal knowledge. We commit to regularly refreshing the automatically generated splits of LIVENEWSBENCH for the foreseeable future, and the human-verified splits for at least two years, contingent on community interest and adoption. This ongoing update process helps maintain the benchmark’s resistance to training-data leakage, preserving its relevance and long-term value.

Multi-hop and challenging benchmark. Questions in LIVENEWSBENCH are constructed by referencing multiple news articles and sources on the same event, and are generated and filtered using rubric-based prompting to ensure sufficient difficulty and objectivity. Consequently, answering a typical question often requires multiple search queries, page visits, and intermediate reasoning steps. This structure makes LIVENEWSBENCH well suited for evaluating agentic web search behavior. In practice, we observe a wide spread in performance across evaluated systems, with accuracy ranging from approximately 10% to 90% depending on the model, agentic framework, and search budget. This variability indicates that LIVENEWSBENCH provides high quality questions with strong discriminative power for current web search LLM agents.

2. Related Work

2.1. Regularly Updated “Live” Benchmarks for LLMs

Evaluating LLMs while limiting data contamination is challenging given LLMs’ massive pretraining datasets. [White et al. \(2025\)](#) introduce LiveBench, a regularly updated benchmark built from newly released math, programming, and general reasoning problems that lie beyond the models’ training cutoffs. [Jain et al. \(2025\)](#) present LiveCodeBench, applying the same idea to coding via recently released competitive-programming problems. MathArena ([Balunović et al., 2025](#)) similarly evaluates math capability using recent math competition problems.

2.2. Benchmarks for Agentic Search Evaluation

Academic reasoning benchmarks as proxies for agentic search. Academic reasoning benchmarks such as Humanity’s Last Exam ([Humanity’s Last Exam Team, 2025](#)) and GAIA ([Mialon et al., 2024](#)) are often used to assess agentic search capabilities ([OpenAI, 2025b](#); [xAI, 2025](#); [DeepSeek-AI, 2025b](#)). However, these benchmarks are primarily designed to measure domain knowledge and STEM reasoning rather than search itself. As a result, they do not inherently isolate or stress web search behavior. For example, enabling search in state-of-the-art models such as GPT-5 yields only a moderate improvement on HLE (from 24.8% to 30.7%), suggesting that search contributes limited additional signal under this evaluation setting.

Factual QA benchmarks for agentic search evaluation. Several factual QA benchmarks have been used to evaluate agentic search capabilities of LLMs ([OpenAI, 2025b](#); [DeepSeek-AI, 2025b](#); [Anthropic, 2025](#); [xAI, 2025](#); [Kimi Team, 2025](#); [GLM-4.5 Team, 2025](#); [Jin et al., 2025](#)). Widely used benchmarks such as SimpleQA ([Wei et al., 2024](#)), BrowseComp ([Wei et al., 2025](#)), DeepSearchQA ([Gupta et al., 2026](#)), TriviaQA ([Joshi et al., 2017](#)), and NaturalQuestions ([Kwiatkowski et al., 2019](#)) are static. As models’ world knowledge improves and training data cutoffs become more recent, an increasing fraction of questions can be answered via memorization alone, reducing their effectiveness for evaluating search. For instance, state-of-the-art models achieve 62.5% accuracy on SimpleQA ([OpenAI](#)) and 82.9% on TriviaQA ([DeepSeek-AI, 2025a](#)) without Internet access, making these benchmarks poorly suited for isolating search capability.

To address the limitations of static benchmarks, several works propose time-sensitive factual QA benchmarks. However, their designs remain vulnerable to memorization. FreshQA ([Vu et al., 2024](#)) and SealQA ([Pham et al., 2025](#)) include fixed questions with evolving answers, but these answers often change slowly and predictably. As discussed in Section 5.1 and Table 2, this static question formula-

Benchmark	Question Updates?	Answer Updates?	Memorization Limited?	Automated Q&A Pair Creation?	Objective & Factual Answers / Evaluation?	Sample Size
BrowseComp (Wei et al., 2025)	No	No	No	No	Yes	1,000
FreshQA (Vu et al., 2024)	No	Partial	No	No	Yes	600
SealQA (Pham et al., 2025)	No	Partial	No	No	Yes	~500
RealTimeQA (Kasai et al., 2023)	Yes	Yes	Yes	No	Yes	~10 / Release
DeepResearch Bench (Du et al., 2025)	No	No Ground Truth	No	No	No	100
LiveResearchBench (Wang et al., 2025)	No	No Ground Truth	Yes	No	No	100
LIVENEWSBENCH (ours)	Yes	Yes	Yes	Yes	Yes	>1K Total / Release

Table 1. Comparison of agentic search and research benchmarks across key design dimensions. We contrast benchmarks by their ability to support regular question and answer updates, limit memorization, scale via automated Q&A generation, and provide objective, factual evaluation. LIVENEWSBENCH is the only benchmark that simultaneously satisfies all these properties while maintaining a large and regularly refreshed sample size.

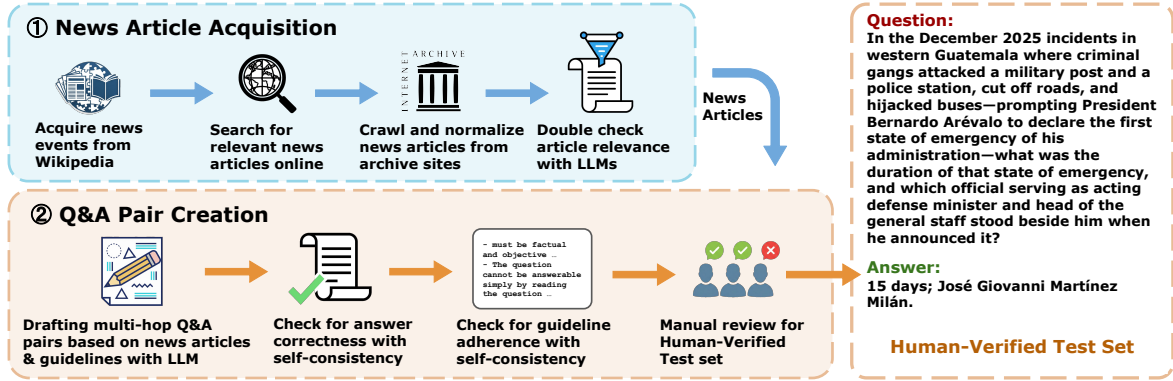


Figure 1. Our automated dataset construction and one example from the Human-Verified Test Set. Our dataset construction pipeline comprises two main components: (1) retrieving news articles from online sources and (2) generating Q&A pairs from the retrieved content. Questions are designed to be challenging, requiring multiple searches, page visits, and reasoning steps.

tion, combined with relatively simple question structures, allows strong LLMs to correctly answer a large portion of questions without search, making these benchmarks less ideal for search evaluation. Moreover, these benchmarks rely on manual question and answer creation, constraining both scale and update frequency. RealTimeQA (Kasai et al., 2023) introduces regularly refreshed questions, but sources its Q&A pairs from manually curated trivia on news websites, resulting in a very small dataset (approximately 10 samples per release).

Deep research benchmarks. Following the introduction of LLM-based deep research agents (OpenAI, 2025c), several benchmarks have been proposed, including DeepResearch Bench (Du et al., 2025), Finder (Zhang et al., 2025), and LiveResearchBench (Wang et al., 2025). These benchmarks typically evaluate long-form reports with rubric-based criteria, which are useful for open-ended research but often lack concise, verifiable ground truth. LIVENEWSBENCH instead targets direct factual questions with short answers, making objective evaluation and RLVR training more straightforward.

3. Dataset Construction

Table 1 summarizes key design properties of representative benchmarks used for evaluating LLM search and research capabilities. As shown, LIVENEWSBENCH targets a different area from prior benchmarks by jointly supporting frequent updates to both questions and answers, limiting memorization, enabling large-scale automated data generation, and retaining objective, verifiable evaluation.

As illustrated in Figure 1, our automated dataset construction process comprises two main components: (1) retrieving major news events from Wikipedia and news articles relevant to the event, and (2) generating Q&A pairs from these articles, followed by both automated and manual validation. We describe each step in detail below.

3.1. Retrieving News Articles

Retrieving seed news events. We begin by collecting a set of major global news events, which serve as the basis for article retrieval. Specifically, we use the [Wikipedia Current Events Archive](#), which provides summaries of high-impact news events worldwide. This allows us to cluster the news articles by major events and later enable us to create Q&A pairs that require searching for multiple sources to answer.

Acquiring URLs of related news articles. For each event, we prompt GPT-4.1 (OpenAI, 2025a) to rewrite the Wikipedia event summary into a search query using a few-shot prompt (provided in Appendix A.4). These queries are submitted to the Brave Search API to retrieve URLs of relevant news articles. To improve factual reliability, we adopt an allowlist of approximately 100 reputable news outlets spanning diverse regions, perspectives, and political leanings (see Appendix A.5 for the full list). To improve temporal relevance, we restrict search results to a 14-day window centered on the event date: 3 days before and 11 days after the date listed on Wikipedia. We combine URLs retrieved from Brave Search and news URLs cited in the Wikipedia Current Events Archive, resulting in a cluster of articles for each news event. In the next step, we retrieve and extract the full text of news articles in each cluster.

Retrieving and extracting news articles. Once we have the URLs, we download the corresponding articles via third-party news archiving services such as archive.today. We use archived versions to ensure long-term stability and reproducibility.

Verifying article relevance. Despite earlier filtering on search results, some retrieved articles may still be unrelated to the intended event due to inaccuracies from the Brave Search API. We therefore use GPT-4.1 (OpenAI, 2025a) to assess the relevance of each article to its corresponding event summary. This final check ensures the clustering of the news articles is correct. The prompt for this step is available in Appendix A.6. After this filtering step, we have 5.3 news articles per event.

3.2. Question-Answer Pair Generation and Validation

Drafting Question-Answer pairs with LLMs. We use a reasoning LLM to generate Q&A pairs from clusters of news articles. For each cluster, we provide GPT-5.1 Thinking (OpenAI, 2025b) with the articles, a set of guidelines aligned with our desiderata, and illustrative examples. The model is prompted to reason step by step, generate multiple candidate Q&A pairs, and select the one that best satisfies the guidelines. If an article cluster does not contain sufficient information to support a meaningful Q&A pair, GPT-5.1 is instructed to skip it. Below, we summarize the high-level requirements for valid Q&A pairs; the full prompt, including examples, is provided in Appendix A.7.

- The Q&A pair must be derived solely from the content of the provided article cluster, without relying on external knowledge or unstated assumptions.
- The question must be factual, self-contained, and unambiguous, such that it can be understood and answered without access to the original articles.

- The answer must be factual, objective, and concise, typically consisting of a short phrase or a few words.
- The Q&A pair must require information from multiple articles within the same news cluster; to enforce this multi-hop requirement, the generation model must cite all referenced articles and sentences in its explanation.
- Avoid Q&A pairs that can be answered using knowledge available prior to January 2025, have ambiguous or subjective answers, or whose ground-truth answers may change over time.

Verifying answer correctness. To ensure the correctness of generated answers, we employ a self-consistency filtering procedure. For each question, we provide the question and its associated source article cluster to GPT-5.1 and ask it to independently derive an answer. We retain Q&A pairs only when both answers produced by GPT-5.1 agree according to the evaluation procedure described in Section 4.3. We adopt self-consistency, as opposed to using a different model, because GPT-5.1 demonstrates the strongest reading-comprehension performance in our setting. In contrast, other models often fail to derive correct answers, which would lead to disproportionately filtering out challenging items and ultimately reducing the difficulty and coverage of the dataset.

Validating guideline adherence. Despite careful prompting, GPT-5.1 occasionally fails to fully follow the Q&A construction guidelines. To ensure guideline compliance, we perform an additional self-consistency validation. After correctness filtering, we provide GPT-5.1 with the Q&A pair, together with a paraphrased version of the guidelines and illustrative examples, and instruct it to reason step-by-step when assessing adherence. We intentionally paraphrase the guidelines to increase robustness and reduce the likelihood of the model trivially matching its earlier reasoning. We retain only those Q&A pairs that adhere to all criteria specified in the paraphrased guidelines, which are detailed in Appendix A.9.

Human-Verified Test Set. To ensure benchmark quality, we construct a human-verified subset of the test set. A human annotator (one of the authors) reviews each Q&A pair for compliance with the guidelines and consistency with the source article. The annotator is only allowed to accept or reject Q&A pairs, and cannot edit them. This process is repeated until 200 validated samples are collected. This subset serves as our high-confidence test benchmark. In practice, we observe that the human annotator rejects $\sim 15\%$ of the Q&A pairs that pass all automated verification.

To assess the clarity of the guideline and the reliability of human annotation, we conduct an additional evaluation in which an NLP researcher unaffiliated with the project inde-

pendently verified 100 annotations. The invited annotator performed verification based on the guidelines shown in Appendix A.9. This annotator rejected 19% of the samples and achieved a 92% overall agreement rate with the original annotator. These results indicate that the guidelines are sufficiently clear for independent researchers to follow and that the resulting human-verified annotations are of high quality.

Examples from our LIVENEWSBENCH dataset are provided in Figure 1 and Appendix A.3.

3.3. Dataset Partitioning and Statistics

We split the dataset into training, validation, and test sets (including the human-verified subset) based on the dates of the underlying news events. Q&A pairs about events from the past two months form the test set, those from the third preceding month form the validation set, and older events are used for training. This chronological ordering ensures the test sets have the lowest likelihood of memorization compared to other sets.

We commit to update the training, validation, and test set of LIVENEWSBENCH quarterly for the foreseeable future. We also commit to update the human-verified test set quarterly for at least two years, depending on community interest.

In the current release, the training set includes over 600 Q&A pairs based on news events from May to September 2025. The validation set contains 170 samples from October 2025, and the test set comprises 340 samples from November and December 2025. From the test set, we construct a human-verified subset of 200 samples via manual review.

To ensure evaluation reliability and reduce costs, all reported results in this paper, unless noted otherwise, are based on the human-verified test set.

The total cost of constructing the release is estimated to be \$700. Details on topic distribution, construction cost, and additional design rationale can be found in Appendix A.2.

4. Experiment Setup

4.1. Agentic Web Search Framework

Although several open-source LLM web search agent frameworks exist (LangChain, 2025; Roucher et al., 2025), we found them unsuitable for our task. These frameworks are typically designed to generate comprehensive deep research reports rather than answering factual questions. As a result, they often consume excessive tokens and search API calls, leading to high evaluation costs and prolonged runtimes.

To address these limitations, we implemented a custom web search agent framework tailored for evaluating LLMs on fact-based queries. We incorporate ReAct-style prompting (Yao et al., 2023), instructing LLMs to provide reason-

ing steps before executing any action. We do not attempt to disentangle reasoning capability from search behavior. Prior work has shown that reasoning naturally complements search, and as a result, ReAct-style prompting has become standard in many agentic search systems (OpenAI, 2025c; LangChain, 2025). Moreover, for several SotA models, such as Gemini 3 Pro, Grok 4, and Kimi K2 Thinking, CoT reasoning cannot be disabled, making it infeasible to isolate the contribution of reasoning from search.

The full web search agent prompts are provided in Appendix A.10. Our framework allows one of the following three actions per step:

- **Search.** The LLM can issue one or more search queries to a Google-like web search engine. In the following step, it receives the top 10 results for each query, including the title, URL, and a relevant content snippet. We use the Tavily Search API (Tavily) to retrieve and rank the web search results given the query.
- **Visit.** The LLM may visit one or more pages from previously returned search results by specifying their URLs. We return the full plain-text content, along with its title and URL, for each selected page.
- **Finish&Answer.** Once the LLM determines that it has gathered sufficient information, it may produce a final answer using a designated answer tag.

Our custom agentic web search framework enables a controlled comparison between models. In our standard evaluation configuration, LLMs are allowed up to 5 search queries and 5 full-page visits before generating an answer. These limits are enforced deterministically by the agent implementation; the prompt exposes the remaining budget so that models can plan their tool use, and it encourages using fewer searches when possible because search efficiency is also measured. In Section 5.3, we performed an ablation study on the search budget, demonstrating our question design requires models to perform multi-hop search.

4.2. Evaluating Integrated LLM Search Systems

Our benchmark also supports the evaluation of integrated LLM-based search systems, including both open-source agentic search frameworks and commercial LLM-based search APIs. In this work, we evaluate the performance of three such systems: OpenAI GPT-5.2 Web Search API (OpenAI, 2025c), OpenAI o4-mini Deep Research API (OpenAI, 2025c), and Claude Sonnet 4.5 Search API (Anthropic, 2025).

Model	LIVENEWSBENCH (%)		FreshQA (%)		SealQA-Hard (%)	
	Human-Verified		Overall	Fast-changing	Overall	Fast-changing
	$n=200$		$n=600$	$n=154$	$n=254$	$n=64$
GPT-5.2	21.5		72.2	47.4	31.9	28.1
Gemini 3 Pro	20.5		74.3	46.8	46.5	31.2
GPT-4.1	14.0		65.7	34.4	26.8	21.9
Kimi K2 Thinking	14.0		63.8	35.7	21.3	10.9
Claude 4.5 Sonnet	13.0		70.8	42.9	23.2	20.3
DeepSeek V3.2 (No Thinking)	12.0		55.3	35.7	27.6	20.3
DeepSeek V3.2 Thinking	10.0		61.0	40.9	31.5	20.3

Table 2. **No-internet accuracy on LIVE NEWSBENCH (ours), FreshQA, and SealQA-Hard.** All results are obtained without internet access. Higher accuracy implies greater reliance on memorized knowledge. Frontier models achieve strong performance on FreshQA and SealQA-Hard, including their fast-changing subsets, despite the absence of search. In contrast, accuracy on LIVE NEWSBENCH remains consistently low across models, indicating reduced memorization and stronger suitability for evaluating agentic web search behavior.

Method	Reasoning Model?	Open Model?	Official Search API?	Searches (avg $\pm$ std)	Visits (avg $\pm$ std)	Accuracy (%)
Models on Local Agent Framework						
DeepSeek V3.2 Thinking	✓	✓	✗	3.3 $\pm$ 1.3	2.6 $\pm$ 1.4	84.5
DeepSeek V3.2 (No Thinking)	✗	✓	✗	3.4 $\pm$ 1.2	2.6 $\pm$ 1.3	83.0
Claude Sonnet 4.5	✗	✗	✗	2.9 $\pm$ 1.1	1.3 $\pm$ 1.3	82.0
Grok 4	✓	✗	✗	2.7 $\pm$ 1.3	1.7 $\pm$ 1.7	82.0
GPT-5.2	✓	✗	✗	2.9 $\pm$ 1.1	1.8 $\pm$ 1.3	74.0
Qwen3.5 397B Thinking	✓	✓	✗	3.2 $\pm$ 0.9	1.4 $\pm$ 1.4	74.0
GPT-4.1	✗	✗	✗	1.7 $\pm$ 0.8	0.6 $\pm$ 1.1	72.5
Kimi K2.5	✓	✓	✗	3.2 $\pm$ 1.1	1.2 $\pm$ 1.3	70.5
GPT-5.4	✓	✗	✗	3.0 $\pm$ 1.2	1.0 $\pm$ 1.3	62.0
Qwen3 235B A22B Instruct	✗	✓	✗	1.9 $\pm$ 0.9	0.4 $\pm$ 0.6	61.5
Gemini 3 Pro	✓	✗	✗	3.4 $\pm$ 0.7	0.6 $\pm$ 1.1	60.5
Qwen3 235B A22B Thinking	✓	✓	✗	2.2 $\pm$ 0.8	0.2 $\pm$ 0.7	60.0
Qwen3 8B	✗	✓	✗	1.6 $\pm$ 0.7	0.1 $\pm$ 0.3	49.5
Kimi K2 Thinking	✓	✓	✗	2.9 $\pm$ 1.0	1.1 $\pm$ 1.3	48.0
GPT-OSS-120B	✓	✓	✗	2.7 $\pm$ 1.2	1.6 $\pm$ 1.0	27.0
Llama 3.1 8B	✗	✓	✗	3.9 $\pm$ 1.1	0.3 $\pm$ 1.0	11.0
Official Agentic Web Search / Deep Research APIs *						
GPT-5.2 Official Web Search API	✓	✗	✓	N/A	N/A	90.0
o4-mini Deep Research API	✓	✗	✓	N/A	N/A	57.0
Claude Sonnet 4.5 Official Web Search API	✗	✗	✓	N/A	N/A	40.0

Table 3. **Human-verified test set results on LIVE NEWSBENCH under the standard evaluation setting** (up to 5 search queries and 5 web page visits per question). * The search budgets for official search and deep research APIs are not controlled, and thus not a fair, controlled comparison to models evaluated on our local agent frameworks. Overall accuracy spans a wide range (11.0%–90.0%), indicating that LIVE NEWSBENCH exhibits a high performance ceiling and strong discriminative power across models and search frameworks.

4.3. Judging Search Outputs

To evaluate LLM-generated answers, we use GPT-4.1 (OpenAI, 2025a) to compare model outputs against ground truth answers. We adopt the evaluation prompt from SimpleQA (Wei et al., 2024), as both our benchmark and SimpleQA use concise phrases as ground-truth answers. We use accuracy as our primary evaluation metric. The evaluation prompt is provided in Appendix A.11.

5. Results and Analysis

5.1. Comparison with other Time-Sensitive Factual QA Benchmarks

To empirically assess the extent to which memorization affects prior time-sensitive benchmarks, we evaluate state-of-the-art LLMs on FreshQA (latest version) and SealQA-

Hard in an offline setting without internet access. We chose SealQA-Hard from SealQA because our dataset does not include adversarial filtering against SotA models, and therefore is more comparable to SealQA-Hard than to SealQA-0. We report results on both the full datasets and their fast-changing subsets. As shown in Table 2, current models achieve strong performance on FreshQA and SealQA-Hard purely from internal knowledge, whereas performance on LIVE NEWSBENCH remains substantially lower. Most notably, Gemini 3 Pro attains 74.3% accuracy on FreshQA and 46.5% on SealQA-Hard without search, but only 20.5% on LIVE NEWSBENCH. These results indicate that existing time-sensitive benchmarks can often be solved through memorization alone, even on subsets intended to emphasize recent information. In contrast, the consistently low no-internet accuracy on LIVE NEWSBENCH suggests stronger resistance to memorization, motivating the need for regu-

Model	Reasoning Model?	Open Model?	Search Budget = 1	Search Budget = 3	Search Budget = 5 (Default)	Search Budget = 7	Improvement (1→7) (%)
DeepSeek V3.2 Thinking	✓	✓	48.5	80.5	84.5	84.5	+36.0
DeepSeek V3.2 (No Thinking)	✗	✓	20.0	79.0	83.0	84.5	+64.5
Claude Sonnet 4.5	✗	✗	53.5	79.0	82.0	67.0	+13.5
GPT-5.2	✓	✗	62.5	72.5	74.0	74.5	+12.0
GPT-4.1	✗	✗	62.5	69.0	72.5	66.5	+4.0
Gemini 3 Pro	✓	✗	29.5	70.0	60.5	74.5	+45.0
Qwen3 8B	✗	✓	22.0	48.0	49.5	49.5	+27.5
Kimi K2 Thinking	✓	✓	7.5	47.0	48.0	52.0	+44.5
GPT-OSS-120B	✓	✓	6.0	28.5	27.0	29.5	+23.5

Table 4. **LIVENEWSBENCH human-verified test set results under varying maximum search budgets.** Rows are sorted by Search Budget = 5 (our default configuration). Increasing the search budget from 1 to 7 consistently improves performance across all LLMs, with gains ranging from 4% to 64.5% (absolute), highlighting the multi-hop nature of our dataset.

Model	Default Acc.	Extended Acc.	Searches	Visits	Correct		Incorrect		≥ 10 Searches	≥ 10 Visits
					Searches	Visits	Searches	Visits		
DeepSeek V3.2 Thinking	84.5	84.0	4.5 $\pm$ 2.9	2.8 $\pm$ 2.0	4.2	2.7	6.1	3.7	7.5	0.5
Claude Sonnet 4.5	82.0	81.0	4.0 $\pm$ 2.4	1.7 $\pm$ 1.5	3.9	1.5	4.5	2.6	6.0	0.0
Tongyi Deep Research	N/A	34.5	2.2 $\pm$ 2.3	0.6 $\pm$ 0.9	3.5	0.8	1.5	0.4	2.0	0.0

Table 5. **Extended tool-budget results on the LIVENEWSBENCH human-verified test set.** The extended setting allows up to 15 searches and 15 full-page visits, compared with the default budget of 5 searches and 5 visits. Accuracy and ≥ 10 tool-use rates are reported in percentages; searches and visits are reported as average $\pm$ standard deviation. Models are unable to make use of extra tool calling quotas, even on harder questions.

larly refreshed benchmarks with complex, event-grounded questions to meaningfully evaluate agentic web search capabilities.

5.2. LIVENEWSBENCH Human-Verified Test Set Results

We evaluate 19 LLMs and three agentic web search APIs on the LIVENEWSBENCH human-verified test set (Table 3). All evaluated LLMs have training-data cutoffs no later than September 2025, and since our test questions are drawn from November-December 2025 events, direct training contamination is unlikely. The results reveal a wide performance spectrum, with accuracy ranging from 11% to 90%. Below, we summarize our key findings.

Large accuracy disparities exist across LLM-based web search agents. We observe substantial variation in accuracy across both models and implementations. The best-performing implementation achieves 90% accuracy, indicating that the dataset contains a low amount of incorrect or invalid Q&A pairs. In contrast, the worst-performing implementation scores as low as 10%, demonstrating that the benchmark is challenging and exhibits strong discriminative power for different agentic web search implementations.

Newer models do not easily saturate LIVENEWSBENCH. To assess whether LIVENEWSBENCH remains challenging for the latest frontier systems, we evaluated three models released after our January 2026 benchmark release: GPT-5.4 Thinking, Qwen3.5-397B Thinking, and Kimi K2.5 Think-

ing. These models achieve scores of 62.0%, 74.0%, and 70.5% respectively, representing changes of -12.0, +14.0, and +22.5 percentage points relative to GPT-5.2, Qwen3-235B Thinking, and Kimi K2 Thinking. All three recent models fall short of the prior state-of-the-art (DeepSeek V3.2 Thinking), and the mixed trend across models suggests that general capability improvements do not reliably translate to gains on LIVENEWSBENCH, indicating the benchmark remains genuinely challenging.

Tool-calling capability significantly affects search performance. By examining execution logs from our local web search agent, we find that some models struggle to follow tool-calling instructions. For instance, Kimi K2 Thinking fails to invoke search or browsing actions using the correct format in 44% of samples, whereas leading models such as Claude Sonnet 4.5 and DeepSeek V3.2 Thinking exhibit failure rates of only 0.5%. These discrepancies in tool-calling reliability help explain the observed performance differences across models.

Search agent implementations can affect performance. Comparing models such as GPT-5.2 and Claude Sonnet 4.5 under our agentic framework with their respective official search agent API, we observe mixed outcomes. GPT-5.2 achieves a 16% accuracy improvement when using its official web search API, while Claude Sonnet 4.5 suffers a 42% accuracy drop under its official API. This suggests that search agent implementations from the model developer may not always result in the best performance.

Because these agentic search APIs offer limited technical

Model	Reasoning Model?	Open Model?	Search Budget = 5 (Default)	No Search	Δ Accuracy (No Search – Search) (%)
DeepSeek V3.2 Thinking	✓	✓	84.5	10.0	-74.5
DeepSeek V3.2 (No Thinking)	✗	✓	83.0	12.0	-71.0
Claude Sonnet 4.5	✗	✗	82.0	13.0	-69.0
GPT-5.2	✓	✗	74.0	21.5	-52.5
GPT-4.1	✗	✗	72.5	14.0	-58.5
Gemini 3 Pro	✓	✗	60.5	20.5	-40.0
Qwen3 8B	✗	✓	49.5	3.5	-46.0
Kimi K2 Thinking	✓	✓	48.0	14.0	-34.0
GPT-OSS-120B	✓	✓	27.0	10.0	-17.0

Table 6. **LIVENEWSBENCH demonstrates limited contamination from training data memorization.** when LLMs are evaluated without Internet access and must rely solely on their internal knowledge. Results with search are copied from Table 3. While state-of-the-art LLMs are capable of generating plausible answers using their world knowledge and reasoning abilities, web search is critical for high performance on this benchmark, as disabling search leads to a 17% to 74.5% (absolute) accuracy drop on this benchmark.

Model	Reasoning Model?	Open Model?	Searches (avg $\pm$ std)	Visits (avg $\pm$ std)	Accuracy (%)	Δ Accuracy (vs Human-Verified Test)
Claude Sonnet 4.5	✗	✗	2.9 $\pm$ 1.1	1.3 $\pm$ 1.2	83.0	+1.0
GPT-5.2	✓	✗	2.8 $\pm$ 1.1	1.8 $\pm$ 1.3	72.0	-2.0
GPT-4.1	✗	✗	1.7 $\pm$ 0.8	0.5 $\pm$ 1.0	69.1	-3.4
Gemini 3 Pro	✓	✗	3.4 $\pm$ 0.8	0.6 $\pm$ 1.1	65.6	+5.1
Kimi K2 Thinking	✓	✓	3.0 $\pm$ 1.0	1.3 $\pm$ 1.4	45.7	-2.3
GPT-OSS-120B	✓	✓	2.7 $\pm$ 1.3	1.5 $\pm$ 1.0	26.3	-0.7

Table 7. **LIVENEWSBENCH full test set results under our standard configuration** (search budget=5). Even with no human verification, results are similar to the human-verified test set: differences in accuracy and search behavior are small, and model rankings remain consistent.

transparency, we can only speculate about the causes of these differences. Plausible factors include differences in prompting strategies, search budgets, underlying search engines, and the model’s familiarity with the provided searching/browsing tools.

Models rely more heavily on search result snippets than full-page visits. Across all models, the average number of search queries exceeds the number of full-page visits. This pattern suggests that models preferentially extract information from search result snippets whenever possible, rather than navigating to and processing full webpages.

5.3. More Searches Help, But Leading Models Plateau Beyond Their Default Budget

In Table 4, we conduct an ablation study varying the maximum number of searches permitted per question to assess its impact on LIVENEWSBENCH performance. The number of allowed page visits is fixed at 5, as we observed that LLMs rely more heavily on search actions than on full-page visits in Section 5.2. Increasing the search budget from 1 to 7 yields consistent performance improvements across all evaluated models, confirming the multi-hop nature of LIVENEWSBENCH, with gains ranging from 4% to 64.5% depending on the model. Notably, open-source models benefit more from increased budgets: models such as DeepSeek V3.1 and Kimi K2 Thinking frequently violate low-budget constraints (e.g., budget = 1), producing format errors that

cause task failures, which explains both their lower performance at minimal budgets and their substantial gains as the budget grows.

Despite these gains, larger tool budgets do not yield proportional improvements for leading models. In an extended-budget probe with 15 searches and 15 full-page visits (Table 5), we evaluate DeepSeek V3.2 Thinking and Claude Sonnet 4.5 as the top-performing models under our local agent framework, alongside Tongyi Deep Research, an open deep research model whose search and page visit traces are fully observable. All three models achieve accuracy comparable to their default-budget scores while consuming on average only a third of their available tool-call quota. For all three models, correct and incorrect responses show similarly modest tool usage (Table 5), indicating that models do not effectively exploit the additional budget on harder questions. Taken together, these results suggest that the bottleneck for current models on LIVENEWSBENCH is not the size of the tool budget but rather the quality of multi-hop reasoning and the ability to issue effective follow-up queries.

5.4. LIVENEWSBENCH Exhibits Limited Memorization

In Table 6, we assess the extent of information contamination in LIVENEWSBENCH by evaluating LLMs on the same human-verified test set, but with Internet access disabled. This setup forces models to rely solely on their internal knowledge and reasoning, allowing us to isolate the con-

tribution of web search. Across all models, we observe substantial reductions in answer accuracy, with accuracy drop ranging from 17.0% to 74.5% (absolute) across different models. This result underscores both the necessity of web access and the contamination-limited nature of our proposed benchmark.

Despite the freshness of the questions and their placement well beyond typical model training cutoffs, we acknowledge that SotA models are still able to correctly answer a subset of questions without Internet access, reaching accuracy between 3.5% and 21.5%. We manually examined correct offline responses from leading models. We found that these models occasionally arrive at correct answers through reasoning based on their internal world knowledge.

For example, when asked: “Which U.S. agency carried out a strike on a Venezuelan dock alleged to load drug-smuggling boats in September 2025?”, many leading models, including GPT-5.2, correctly inferred that *the CIA* likely carried out the strike, given its long history of operating covertly abroad. This example underscores that, despite our deliberate efforts to design LIVENEWSBENCH with fresh news events to minimize memorization, state-of-the-art models still exhibit strong world knowledge and reasoning capabilities, making it inevitable that they can correctly guess a nontrivial portion of such questions.

5.5. Human-Verified Test Set vs. Full Test Set

Table 7 reports results of six LLMs on the full LIVENEWSBENCH test set. We limit evaluation to six models to reduce cost, as the full test set is larger and more expensive to run.

We observe that accuracy differences between the two sets are minimal across all models. The largest gap is seen with Gemini 3 Pro, which achieves 60.5% accuracy on the human-verified set and 65.6% on the full set (a 5.1% increase). Search and page visit counts are also consistent, with no model exceeding a delta of 0.1 per question on average. Model rankings remain consistent across all models compared.

These findings suggest that our automated Q&A generation pipeline is robust, producing high-quality questions and answers. This is an extrinsic evaluation of the fully automated test set resulting in the same ranking as on the human validated set. Even if some quality degradation exists in the full set due to the lack of human filtering, its impact on both sample quality and overall benchmark outcomes is limited.

6. Conclusion And Future Work

We introduced LIVENEWSBENCH, a regularly updated benchmark for evaluating LLMs’ agentic web search capabilities. By automatically generating fresh Q&A pairs

from recent news, LIVENEWSBENCH allows for frequent updates, reduces contamination, and separates models’ internal knowledge from search behavior. Its multi-step design and human-verified subsets enable rigorous and reliable evaluation of models’ agentic web search capabilities.

For future work, we are working on a longitudinal extension that targets temporally sensitive questions whose correct answers may change as events unfold. This paper does not study that setting, and the benchmark and pipeline described here intentionally focus on questions with non-shifting ground-truth answers to facilitate evaluation.

Impact Statement

This work advances the evaluation of LLMs with agentic web search capabilities. We introduce LIVENEWSBENCH, a rigorously constructed and regularly updated benchmark that automatically generates fresh question-answer pairs from recent news, encouraging evaluation on information beyond an LLM’s training data. By requiring multi-step querying, page visits, and reasoning, LIVENEWSBENCH supports clearer measurement of end-to-end search behavior and helps distinguish internal model knowledge from retrieved evidence. We expect the benchmark, public leaderboard, and released code to enable more reproducible comparisons across systems and to accelerate progress on reliable and equitable information access. In addition, our automated curation pipeline supports frequent updates and provides open, verifiable data that can help address the scarcity of training resources for agentic web search models.

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A. Appendix

A.1. LLM Inference Settings

Model	Inference Provider	Temp	Top_p	Max tokens to generate
Claude Sonnet 4.5	Anthropic	1.0	0.95	4096
Gemini 3 Pro	Google	1.0	0.95	16384
DeepSeek V3.2 Thinking	Fireworks	1.0	0.95	16384
DeepSeek V3.2 (No Thinking)	Fireworks	1.0	0.95	4096
GPT-OSS-120B	Fireworks	1.0	1.0	16384
Kimi K2 Thinking	Fireworks	1.0	1.0	16384
Llama 3.1 8B	Fireworks	0.6	0.9	4096
Qwen3 235B A22B Instruct	Fireworks	0.7	0.8	4096
Qwen3 235B A22B Thinking	Fireworks	0.6	0.95	16384
Qwen3 8B	Fireworks	0.7	0.8	4096
GPT-4.1	OpenAI	0.7	0.95	4096
GPT-5.2	OpenAI	1.0	1.0	16384
Grok 4	xAI	1.0	1.0	16384

Table 8. Inference configuration used by the evaluation script. We always set the max token to generate to 4096 for non-thinking models, and 16384 for thinking models. Temperature and Top_P are set to each model’s recommended settings.

A.2. Additional Details and Statistics on LIVENEWSBENCH Dataset Construction

A.2.1. LIVENEWSBENCH TOPIC DISTRIBUTION

Table 9 reports the topic distribution of the human-verified test set, using categories from Wikipedia Current Events. Table 10 reports the geographic distribution, assessed by GPT-4.1. All answers are concise, objective, and factual, typically consisting of 1–5 words. We do not provide a difficulty breakdown, as difficulty is inherently subjective and difficult to define in a principled way.

Topic	(%)
Armed conflicts and attacks	24.5
Politics and elections	20.0
Law and crime	19.5
International relations	10.5
Disasters and accidents	9.0
Business and economy	7.0
Arts and culture	3.5
Sports	3.5
Health and environment	2.0
Science and technology	0.5

Table 9. Topic distribution of the LIVENEWSBENCH human-verified test set.

Region	(%)
Asia	32.5
Europe	22.5
North America	17.5
Africa	12.5
South America	7.5
Multiple continents	5.5
Oceania	2.0

Table 10. Geographic distribution of the LIVENEWSBENCH human-verified test set.

A.2.2. ADDITIONAL DETAILS ON LIVENEWSBENCH CONSTRUCTION COST

All of the reported 700 USD data curation cost reported in the paper stems from LLM API calls: approximately 100 USD was spent on verifying article relevance to news events, 200 USD on drafting candidate Q&A pairs, and 400 USD on subsequent correctness and guideline adherence verification steps. Human verification is performed by students conducting for-credit research at our institution, totaling 5-10 hours per release. Under our institutional policy, students engaged in for-credit research cannot receive monetary compensation, so no cost is attributed to human verification.

A.2.3. CONSIDERATIONS ON ADOPTING GPT-5.1 FOR QUESTION GENERATION AND VALIDATION

During the experiment design phase, we evaluated multiple closed and open-source models, including the GPT-5 series, Claude Sonnet 4.5, Kimi K2 Thinking, and DeepSeek V3.2. Qualitative human assessment determined that the GPT-5 series was most effective at generating challenging, multi-hop questions suited to evaluating state-of-the-art LLM agentic web search capabilities; other models tended to produce simpler, more straightforward questions that inflate benchmark performance. Since we also intend to evaluate the latest GPT-5 series model (GPT-5.2) and while avoiding self-evaluation bias, we chose to use GPT-5.1 to construct and validate our benchmark questions. It is worth noting that LIVENEWSBENCH scores are designed to be compared within a single release, so it is possible to adopt newer, more capable models for LIVENEWSBENCH QA pair generation and validation in future releases.

A.2.4. PRACTICALITY OF ADOPTING LIVENEWSBENCH FOR RLVR TRAINING

LIVENEWSBENCH features $\sim 1K$ factual QA questions where the ground truth answers are short phrases / sentences, making the dataset suitable for training web search agentic with RLVR. However, training agentic web search models comes with a high cost: even under conservative assumptions: 50 RL steps, 256 samples per step, 16 rollouts per sample, an average of 3 searches and 3 full-page visits per rollout, and a 32K context window, a single training run would require approximately 200K rollouts, 1.2M search API calls, and 6.4B tokens. At standard search API pricing (0.008 USD per call, e.g., Tavily), search costs alone would reach 10,000 USD. Training a 7-8B parameter model would further require 750 H200 GPU hours (2,000 USD), bringing the total cost per run to approximately 12,000 USD.

For these reasons, we focused our contribution on the data acquisition pipeline, automated Q&A construction and verification, and standardized evaluation metrics, and we leave RLVR training to the broader community and resource-rich research groups. We hope our dataset and pipeline serve as a useful foundation for such future work.

A.2.5. LIVENEWSBENCH HUMAN BASELINE PERFORMANCE

We conducted a preliminary human baseline study. A human NLP researcher (not an author) attempted 50 questions from the human-verified test set under identical constraints: up to 5 web searches and 5 full-page visits, using Google Search with all LLM-assisted features disabled.

The human participant achieved a 50% accuracy, using an average of 2.4 searches and 3.4 full page visits per question. The 50% human accuracy falls between Qwen3 235B A22B Thinking (9th, 60.0%) and Qwen3 8B (10th, 49.5%), well below the SotA LLMs. These results are unsurprising for two reasons. First, LLMs routinely exploit advanced search syntax, such as site: or quoted phrases to enforce must-include keywords, yielding more precise queries. Second, SotA LLMs are explicitly trained for agentic search via RLVR, and models optimized in this way are known to often match or exceed average human performance on well-defined, verifiable tasks. It is therefore expected that LLMs outperform a typical human researcher on this benchmark.

A.3. Additional LIVENEWSBENCH Human Verified Examples

Q: According to 2025 news coverage of China’s provisional antisubsidy tariffs on dairy imports from the European Union, which specific corporate entities of a Dutch dairy conglomerate from an EU member state that had voted in favour of higher EU tariffs on Chinese-made electric vehicles were reported as being subject to the highest provisional duty rate reserved for exporters that did not participate in the investigation, and what was that duty rate?

A: FrieslandCampina Belgium NV and FrieslandCampina Nederland BV, at 42.7%.

Q: For the 83rd Golden Globes, organizers introduced a new Best Podcast category with an eligibility window defined by a specific date range for episode releases. Using the final date of that eligibility window and the calendar date on which the 83rd Golden Globe nominations were announced, how many days after the eligibility window closed were the nominations made public?

A: 69 days.

A.4. Search Keyword Generation Prompt

You are given a list of headlines and summaries for news events. Your task is to write Google search queries based on the headlines and the summaries. Make sure the queries that you write are concise and relevant, and follows the best practices of using Google search. Provide your answers directly, and do not say anything else. Now, read the first seven examples carefully, and finish the last sample in the same manner.

Headline: 2022 monkeypox outbreak - 2022 monkeypox outbreak in Europe - 2022 monkeypox outbreak in Germany
Summary: Germany reports almost one hundred new cases of monkeypox.
Query: Germany monkeypox outbreak new cases

Headline: Mianwali air base attack
Summary: Nine Tehreek-e-Jihad jihadists are killed during a shootout with soldiers when they tried to storm a training air base in Mianwali, Punjab, Pakistan.
Query: Mianwali air base attack

Headline: LGBT rights in Thailand, Recognition of same-sex unions in Thailand
Summary: The Senate of Thailand passes a marriage equality bill that will legalize same-sex marriage in the country, with the bill now awaiting royal assent.
Query: Thailand's Senate passes same-sex marriage bill

Headline:
Summary: Thousands of people, including teachers and students, protest across Hungary against the government of Viktor Orban, demanding higher salaries and the right to strike amid a high level of inflation in the country.
Query: Hungary inflation protest

Headline: War in Sudan
Summary: Residents of White Nile State flee south toward the border with South Sudan amid rumours of an impending Rapid Support Forces assault on the region.
Query: Nile State residents flee war in Sudan

Headline: Colombian conflict
Summary: Former Colombian National Army General Mario Montoya Uribe is charged for his role in the "false positives" scandal. Montoya is accused of carrying out extrajudicial executions of 130 people.
Query: Mario Montoya Uribe charged Colombia

Category: Politics and elections
Headline: Doug Burgum 2024 presidential campaign
Summary: Governor of North Dakota Doug Burgum announces his candidacy for President of the United States in 2024.
Query: Doug Burgum presidential campaign announcement

Headline: your headline
Summary: your summary
Query:

A.5. Allowed News Outlets

```
*.altnet.org/*, *.apnews.com/*, *.theatlantic.com/*, *.thedailybeast.com/*, *.
democracynow.org/*, *.huffpost.com/*, *.theintercept.com/*, *.jacobin.com/*, *.
motherjones.com/*, *.msnbc.com/*, *.thenation.com/*, *.nytimes.com/*, *.newyorker.
com/*, *.slate.com/*, *.vox.com/*, *.abcnews.go.com/*, *.axios.com/*, *.bloomberg.
com/*, *.cbsnews.com/*, *.cnn.com/*, *.insider.com/*, *.nbcnews.com/*, *.npr.org/*,
*.politico.com/*, *.propublica.org/*, *.semafor.com/*, *.time.com/*, *.usatoday.com
/*, *.washingtonpost.com/*, *.csmonitor.com/*, *.cnbc.com/*, *.forbes.com/*, *.
newsnationnow.com/*, *.newsweek.com/*, *.reason.com/*, *.reuters.com/*, *.wsj.com/*,
*.thedispatch.com/*, *.theepochtimes.com/*, *.foxbusiness.com/*, *.thefp.com/*, *.
justthenews.com/*, *.nationalreview.com/*, *.nypost.com/*, *.upward.news/*, *.
washingtonexaminer.com/*, *.washingtontimes.com/*, *.theamericanconservative.com/*,
*.spectator.org/*, *.theblaze.com/*, *.breitbart.com/*, *.cbn.com/*, *.dailycaller.
com/*, *.dailymail.co.uk/*, *.dailywire.com/*, *.foxnews.com/*, *.thefederalist.com
/*, *.ijr.com/*, *.newsmax.com/*, *.oann.com/*, *.thepostmillennial.com/*, *.
freebeacon.com/*, *.bbc.com/*, *.theguardian.com/*, *.independent.co.uk/*, *.ft.com
/*, *.telegraph.co.uk/*, *.dw.com/*, *.thelocal.de/*, *.spiegel.de/international/*,
*.france24.com/*, *.thelocal.fr/*, *.lemonde.fr/en/*, *.euronews.com/*, *.ukrinform.
net/*, *.kyivindependent.com/*, *.kyivpost.com/*, *.rt.com/*, *.sputniknews.com/*, *.
themoscowtimes.com/*, *.chinadaily.com.cn/*, *.globaltimes.cn/*, *.scmp.com/*, *.
japantimes.co.jp/*, *.asia.nikkei.com/*, *.timesofindia.indiatimes.com/*, *.thehindu.
com/*, *.economictimes.indiatimes.com/*, *.telegraphindia.com/*, *.indianexpress.com
/*, *.aljazeera.com/*, *.haaretz.com/*, *.timesofisrael.com/*, *.jpost.com/*, *.
mexiconewsdaily.com/*
```

A.6. Article Relevance Filtering Prompt

You are presented with a news event summary and an article that is potentially related to the given news event. Your task is to determine if the article is actually relevant to the news event. You should output "Yes" only if the article is relevant, and "No" otherwise. The contents of the article do not need to match exactly - as long as they are reporting roughly the same event as in the news event summary, they are considered relevant. However, articles that contains completely different events (e.g. Gun violence in US in the Article, but the news event is about war in Ukraine) should be marked as irrelevant. Ignore irrelevant sections such as footers, related articles, and ads. You may think step by step, but you must end the response with your final verdict, formatted as: "Answer: Yes" or "Answer: No".

News Event Summary:
{summary}

Article:
{title}
{content}

A.7. Q&A Pair Creation Prompt

Please generate one factual Q&A pairs based solely on a set of news articles on a specific news event, provided below. The goal of the Q&A pairs is to assess the test-taker's ability to search information online and reason over the content. Follow the following instructions carefully:

Q&A Pair Requirements

1. Source of Truth:

- The question and answer must be ****exclusively**** derived from the provided articles below. Do not rely on any external knowledge, assumptions, or information

not present in the text.

- If multiple articles are available below, you **must** create QA pairs that make use of multiple articles provided below, and cannot be answered by just using any single articles below. Using multiple articles to cross-check the same evidence is insufficient; the question must require synthesizing information from multiple articles to arrive at the correct answer. However, you can ignore this requirement if only one article is provided.

2. Question Requirements:

- The question must be factual, self-contained, and unambiguous.
- You must always assume the test-takers do not have access to the articles you are reading right now, and a person reading the question alone (without the articles below) should understand exactly what is being asked. Therefore, problems like "according to the CNN article" are forbidden, because the test-taker may not find the same article as you.
- The question cannot be answerable simply by reading the question itself without any online research or additional articles. For example, you should not cite key statistics within the question - let the test-taker find the info online on their own. You also should not ask any questions that could be inferred with knowledge before march 2025.

3. Answer Requirements:

- The answer must be a factual, objective, and concise statement (a few words or a short phrase) that is explicitly verifiable by using an LLM to check against a ground truth answer.
- Avoid quotes, opinions, and subjective interpretations.
- Do not inflate difficulty by piling up multiple math calculations.

4. Difficulty Requirements:

- The Q&A pairs should be **very challenging** and require multi-hop web search and reasoning to derive the answer.
- The Q&A pairs should be "Google-Proof": it should not be common knowledge or easily discoverable via a single Google search of the question terms.
- However, bear in mind that the test-takers would not be able to access the articles you are reading right now. Instead, they will perform their own online research and reasoning to find the answer. Therefore, you should aim for a question that is challenging but answerable through multiple rounds of online search and reasoning.

5. Temporal & Stability Requirements:

- The Q&A pairs must not be answerable using knowledge available before April 2025.
- Avoid facts that are likely to change over time (e.g., fluctuating statistics, number of victims in an event, as these facts might change as the reporting / events develop).
- Avoid QA pairs where multiple alternative answers may exist.

Good and Bad Examples

- Good Examples:

- Question: During SpaceX's dual moonshot launch on 15 January 2025, how many payloads in total were carried by the two lunar landers released from the Falcon 9 rocket? Answer: 16. Reasons: This Q&A pair is good because it is factual, self-contained, and unambiguous. It does not require access to a specific article to answer the question. It is also Google-Proof (needs multiple searches) and requires multi-hop reasoning.
- Question: In the timeline reported for Argentina's \$Libra cryptocurrency launch, how much time elapsed between KIP Protocol's supportive post on X and the Argentine president's office issuing its statement blaming KIP for the project? Answer: 10 hours and 2 minutes. Reasons: This Q&A pair factual, self-contained, and unambiguous. It could be answered without access to a specific article. The question is clear about events that are referred to, and the time and dates are fixed and therefore do not change over time. It also requires multiple steps of online research, and deriving the answer based on multi-hop reasoning.

- Question: Which former member of Donald Trump's personal legal team, while serving as acting U.S. deputy attorney general, called for an investigation into Sheriff Derek R. Osborne for releasing Jesus Romero-Hernandez from Tompkins County custody? Answer: Emil Bove III. Reasons: The Q&A pair is factual and self-contained. It is unambiguous because the question provides sufficient context to identify the specific individual being asked about. It is also Google-Proof as it requires multiple steps of search and reasoning to get right.

- Bad Examples:

- Question: In reporting on Chelsea's 3-0 win over Paris Saint-Germain in the 2025 FIFA Club World Cup final, one article said Chelsea would receive \$40 million "for the final alone," while another article estimated Chelsea's total prize money from winning the tournament at \$114.6 million. Based on those two reported figures, about how much of Chelsea's prize money came from stages other than the final? Answer: About \$74.6 million. Reasons: This question is bad because it included key information in the question itself, instead of letting test takers to find information online, making it answerable simply by reading the question.

- Question: In July 2025 coverage of Hong Kong police offering a HK\$200,000 bounty per wanted activist, one report converted that HK\$200,000 into U.S. dollars and another converted the same HK\$200,000 into Australian dollars; based on those two conversions, what approximate Australian-dollar-per-U.S.-dollar exchange rate is implied? Answer: About A\$1.53 per US\$1. Reasons: This question is bad because you should avoid questions in the form of "one report said X, another report said Y".

- Question: After Germany announced it would no longer authorize exports of military equipment that could be used in Gaza, what total value of German arms exports to Israel since October 7, 2023 (as cited by Germany's Economy Ministry in a DPA report) was mentioned, and what share of Israel's total arms imports between 2020 and 2024 did Germany account for (according to SIPRI)? Answer: 485 million Euros and 33%. Reasons: This question is bad because it is underspecified and requires access to specific articles to answer. We cannot assume that test-takers will be able to find the exact same articles being referred to.

- Question: According to the containment figures reported in the CNN article, by how many percentage points did the Hurst Fire's containment exceed that of the Palisades Fire? Answer: 81 percentage points. Reasons: This question is bad because it is not self-contained and requires access to a specific article to answer. It is ambiguous which CNN article is being referred to. Similar expressions like "according to the CNN article", "according to one report", "One article (Reuters) provides", etc. should be avoided.

- Question: According to the draft Gaza ceasefire agreement accepted by Hamas in January 2025, if roughly 600 humanitarian aid trucks are to enter the Strip each day during the entire 42-day first phase, how many aid trucks in total are expected to enter Gaza in that phase? Answer: 25,200. Reasons: This question is bad because it can be answered by reading the question itself. Also, as the Gaza conflict is an ongoing event, the ceasefire agreement and its terms may change over time, making the answer potentially unstable.

- Question: In the news report that described Trump's call for a 100% tariff on foreign films one day after meeting Jon Voight at Mar-A-Lago and cited Marvel's Thunderbolts* as mostly U.S.-made but with shoots in Malaysia and a score recorded in London, by how many does the number of foreign countries listed as common filming locations exceed the number of U.S. states named as offering generous tax incentives? Answer: 5. Reasons: Trump's tariff plan is an ongoing event, so the number of countries and states involved may change over time. Also, the question is ambiguous because it was not clear which report is being referred to. Finally, the question is overly complex and convoluted.

- Question: In the late-July 2025 cease-fire talks between Thailand and Cambodia held in Putrajaya, which senior analyst for Southeast Asia suggested that President Trump's tariff threat may have pushed Bangkok to accept mediation, and what organization is he affiliated with? Answer: Matthew Wheeler of the International Crisis Group. Reasons: This question is bad because alternative answers may exist. Other analysts and organization may have suggested similar opinions, so valid alternative answers may exist.

- Question: Based on reporting around Jamaica's September 3, 2025 general election, if Andrew Holness carries out his pledge to double the national minimum

wage while the standard workweek for that wage stays the same length, what would be the approximate hourly minimum wage in U.S. dollars after the increase, using the exchange rate cited in that election coverage? Answer: About \$5.03 per hour. Reasons: This question is bad before one can infer the answer by memorizing the minimum wage and exchange rate in early 2025 before the election, and then guess the answer by doubling that current number. So this is a question is answerable simply by reading the question, without having to search online. Not quite challenging.

Step-by-Step Process

Please approach this task by thinking step by step in your internal reasoning process. First, propose a few candidate Q&A pairs. Then, for each candidate, carefully check its eligibility against the above requirements, compare them to the good and bad examples provided above, and refine the Q&A pairs if needed. You should also think about whether the Q&A pair you selected is stable and unlikely to change over time. Finally, select the best Q&A pair that meets all the criteria. However, if you are unsure you can create valid Q&A pairs that meets all the requirements, feel free to skip this set of articles. It is ****preferred**** to skip than to create a low-quality Q&A pair.

Output Format

Output the final Q&A pair and temporal stability assessment in the following format. Pick the Q&A pair that is the most challenging and yet follows all of the guidelines above.

Question: Your proposed question here

Answer: Your proposed answer here.

Temporal Stability: Your judgement of the temporal stability of the Q&A pair.

Available options: "Unlikely to change over time", "May change in the next few years", "May change within one year".

Explanation: A detailed explanation to justify the correct. At the end of your explanation, please also quote the specific sentences or data points from the articles that support your answer, using the format "Article [X]: '...'" where X is the article number (starting from 1).

If the articles do not contain sufficient information to create a valid Q&A pair that meets all the requirements, you can skip. In this case, output exactly the following:

Question: N/A

Answer: N/A

Temporal Stability: N/A

Explanation: your explanation here.

Now, please read the articles below and generate a list of Q&A pairs.

Articles

{article_content}

A.8. Q&A Pair Correctness Verification Prompt

Please answer the question presented below based on the following article(s). Please think step by step, and put your final answer after "Answer:" at the end of your response, on a separate line, as "Answer: your_answer". Follow the output format strictly. If the question cannot be answered based on the article(s), respond with "Answer: Cannot be determined from the article(s) provided."

Question: {question}

Article

{article}

A.9. Q&A Pair Guidance Adherence Prompt

You are given a question answer pair and a set of criterias and good/bad examples. Your task is to determine if the question answer pair ****meets all the criteria**** listed below by comparing the proposed Q&A pair against the criteria and examples. Please think step by step, and provide your verdict at the end of your response using "Meets all criteria: yes" or "Meets all criteria: no".

Q&A Pair Requirements

1. Question Requirements:

- The question must be factual, self-contained, and unambiguous. It should not rely on access to a specific news article, report, or a particular source. Therefore, problems like "according to the CNN article", "according to the UN report" are forbidden, because the test-taker may not have access to the same article or report.
- The question cannot be answerable simply by reading the question itself without any online research or additional articles. For example, there should not be key statistics within the question - let the test-taker find the info online on their own.
- Reject questions that could be inferred with knowledge before January 2025, or questions where you can guess the answer based on prior knowledge without searching online.

2. Answer Requirements:

- The answer must be a factual, objective, and concise statement (a few words or a short phrase) that is explicitly verifiable by using an LLM to check against a ground truth answer.
- Avoid quotes, opinions, and subjective interpretations.
- Do not inflate difficulty by piling up multiple math calculations.

3. Difficulty Requirements:

- The Q&A pairs should be ****very challenging**** and require multi-hop web search and reasoning to derive the answer.
- The Q&A pairs should be "Google-Proof": it should not be common knowledge or easily discoverable via a single Google search of the question terms.

4. Temporal & Stability Requirements:

- The Q&A pairs must not be answerable using knowledge available before April 2025.
- Avoid facts that are likely to change over time (e.g., fluctuating statistics, number of victims in an event, as these facts might change as the reporting / events develop).
- Avoid QA pairs where multiple alternative answers may exist.

Good and Bad Examples

- Good Examples:

- Question: During SpaceX's dual moonshot launch on 15 January 2025, how many payloads in total were carried by the two lunar landers released from the Falcon 9 rocket? Answer: 16. Reasons: This Q&A pair is good because it is factual, self-contained, and unambiguous. It does not require access to a specific article to answer the question. It is also Google-Proof (needs multiple searches) and requires multi-hop reasoning.
- Question: In the timeline reported for Argentina's \$Libra cryptocurrency launch, how much time elapsed between KIP Protocol's supportive post on X and the Argentine president's office issuing its statement blaming KIP for the project? Answer: 10 hours and 2 minutes. Reasons: This Q&A pair factual, self-contained, and unambiguous. It could be answered without access to a specific article. The question is clear about events that are referred to, and the time and dates are fixed and therefore do not change over time. It also requires multiple steps of online research, and deriving the answer based on multi-hop reasoning.
- Question: Which former member of Donald Trump's personal legal team, while serving as acting U.S. deputy attorney general, called for an investigation into Sheriff Derek R. Osborne for releasing Jesus Romero-Hernandez from Tompkins County custody? Answer: Emil Bove III. Reasons: The Q&A pair is factual and self-contained.

It is unambiguous because the question provides sufficient context to identify the specific individual being asked about. It is also Google-Proof as it requires multiple steps of search and reasoning to get right.

- Bad Examples:

- Question: In reporting on Chelsea's 3-0 win over Paris Saint-Germain in the 2025 FIFA Club World Cup final, one article said Chelsea would receive \$40 million "for the final alone," while another article estimated Chelsea's total prize money from winning the tournament at \$114.6 million. Based on those two reported figures, about how much of Chelsea's prize money came from stages other than the final?

Answer: About \$74.6 million. Reasons: This question is bad because it included key information in the question itself, instead of letting test takers to find information online, making it answerable simply by reading the question.

- Question: In July 2025 coverage of Hong Kong police offering a HK\$200,000 bounty per wanted activist, one report converted that HK\$200,000 into U.S. dollars and another converted the same HK\$200,000 into Australian dollars; based on those two conversions, what approximate Australian-dollar-per-U.S.-dollar exchange rate is implied? Answer: About A\$1.53 per US\$1. Reasons: This question is bad because you should avoid questions in the form of "one report said X, another report said Y".

- Question: After Germany announced it would no longer authorize exports of military equipment that could be used in Gaza, what total value of German arms exports to Israel since October 7, 2023 (as cited by Germany's Economy Ministry in a DPA report) was mentioned, and what share of Israel's total arms imports between 2020 and 2024 did Germany account for (according to SIPRI)? Answer: 485 million Euros and 33%. Reasons: This question is bad because it is underspecified and requires access to specific articles to answer. We cannot assume that test-takers will be able to find the exact same articles being referred to.

- Question: According to the containment figures reported in the CNN article, by how many percentage points did the Hurst Fire's containment exceed that of the Palisades Fire? Answer: 81 percentage points. Reasons: This question is bad because it is not self-contained and requires access to a specific article to answer. It is ambiguous which CNN article is being referred to. Similar expressions like "according to the CNN article", "according to one report", "One article (Reuters) provides", etc. should be avoided.

- Question: According to the draft Gaza ceasefire agreement accepted by Hamas in January 2025, if roughly 600 humanitarian aid trucks are to enter the Strip each day during the entire 42-day first phase, how many aid trucks in total are expected to enter Gaza in that phase? Answer: 25,200. Reasons: This question is bad because it can be answered by reading the question itself. Also, as the Gaza conflict is an ongoing event, the ceasefire agreement and its terms may change over time, making the answer potentially unstable.

- Question: In the news report that described Trump's call for a 100% tariff on foreign films one day after meeting Jon Voight at Mar-A-Lago and cited Marvel's Thunderbolts* as mostly U.S.-made but with shoots in Malaysia and a score recorded in London, by how many does the number of foreign countries listed as common filming locations exceed the number of U.S. states named as offering generous tax incentives? Answer: 5. Reasons: Trump's tariff plan is an ongoing event, so the number of countries and states involved may change over time. Also, the question is ambiguous because it was not clear which report is being referred to. Finally, the question is overly complex and convoluted.

- Question: In the late-July 2025 cease-fire talks between Thailand and Cambodia held in Putrajaya, which senior analyst for Southeast Asia suggested that President Trump's tariff threat may have pushed Bangkok to accept mediation, and what organization is he affiliated with? Answer: Matthew Wheeler of the International Crisis Group. Reasons: This question is bad because alternative answers may exist. Other analysts and organization may have suggested similar opinions, so valid alternative answers may exist.

- Question: Based on reporting around Jamaica's September 3, 2025 general election, if Andrew Holness carries out his pledge to double the national minimum wage while the standard workweek for that wage stays the same length, what would be the approximate hourly minimum wage in U.S. dollars after the increase, using the exchange rate cited in that election coverage? Answer: About \$5.03 per hour. Reasons: This question is bad before one can infer the answer by memorizing the minimum wage

and exchange rate in early 2025 before the election, and then guess the answer by doubling that current number. So this is a question is answerable simply by reading the question, without having to search online. Not quite challenging.

```
# Proposed Q&A Pair to Verify
Question: {question}
Answer: {answer}
```

A.10. Web Search Agent Prompt

You are a web search agent that performs web searches and reasoning to answer user queries. \

You are excellent at carefully following instructions and thinking step by step. \

You are also good at following output format instructions.

Actions you can take

Search

You can use this action to search online via a search engine (e.g. Google). \

You can perform one or multiple searches at a time. However, you should try to minimize the total number of searches needed when possible. \

This is just plain web search, so you should make sure your queries are concise and specific, similar to how you would search online on Google. \

The results, which includes the article title, url, and snippet, will be returned to you in the next turn as a user message. \

You can choose to see the full content of any article by using the "Click" action described later.

To invoke this action, use the following format:

```
<action>
type: Search
your_query_1
your_query_2
your_query_3
...
</action>
```

Click

You can use this action to click on a search result link to see the full content of the article. \

You can click on one or multiple links at a time. However, you should try to minimize the total number of clicks needed when possible. \

The full content of the article will be returned to you in the next turn as a user message. \

You can only click on links that are returned from the Search action in previous turns. Using links from a earlier search result is allowed. \

However, clicking on links that have never been returned to you is not allowed and will be treated as a task failure.

To invoke this action, use the following format:

```
<action>
type: Click
url_1
url_2
url_3
...
</action>
```

Finish

You can use this action to finish the task and provide your final answer. \

The final answer should be a concise and complete answer to the original question,

based on the information you have gathered. \
 The final answer is usually a phrase, so a few words long. \
 You should only use this action when you are confident that you have gathered enough information to answer the question.

To invoke this action, use the following format:

```
<action>
type: Finish
your_final_answer
</action>
```

You must follow the guidelines below exactly
 ## One action block per turn
 You should always invoke one and only one action block per turn. \
 Responses that do not have a action block with one of the three allowed actions types will be considered invalid. \
 You can indeed search and click multiple times in one action, but there should only be one action block in your response.

Think step by step
 Please always approach the task by thinking step by step. Always think about the next action before committing to it.

So in each turn, you should first provide a couple paragraphs of reasoning about the next action you will take, \
 and then invoke the action using:
 <action>
 ...
 </action>.

Minimize total number of actions
 You should try to minimize the total number of actions you take to complete the task. \
 This means you should try to minimize the total number of action blocks you invoke, \
 as well as the total number of searches and clicks needed before reaching the final answer.

Respect the search and click budgets
 You can perform up to {max_searches} searches and {max_clicks} clicks in total.

Each individual search queries and clicks count towards the total budget. So if you invoked search action with 3 queries, \
 that counts as 3 searches towards your total budget. Similarly, if you clicked on 2 links in one click action, \
 that counts as 2 clicks towards your total budget.

When you ran out of both searches and clicks, you must use the Finish action to provide your final answer. \
 Running out of either searches or clicks means you can no longer use the corresponding action, but you can still use the other action.

If you exceed either budget, you will fail the task instantly.

You have used {used_searches} out of {max_searches} searches, and {used_clicks} out of {max_clicks} clicks.

You may use the Search or Click actions if you have remaining budget, or use the Finish action to provide your final answer.

Now, let's think step by step to determine the next action to take.

A.11. LLM Grading Prompt

Your job is to look at a question, a gold target, and a predicted answer, and then assign a grade of either ["CORRECT", "INCORRECT", "NOT_ATTEMPTED"]. First, I will give examples of each grade, and then you will grade a new example.

The following are examples of CORRECT predicted answers.

```

Question: What are the names of Barack Obama's children?

Gold target: Malia Obama and Sasha Obama

Predicted answer 1: sasha and malia obama

Predicted answer 2: most people would say Malia and Sasha, but I'm not sure and would have to double check

Predicted answer 3: Barack Obama has two daughters. Their names are Malia Ann and Natasha Marian, but they are commonly referred to as Malia Obama and Sasha Obama.

Malia was born on July 4, 1998, and Sasha was born on June 10, 2001.

```

These predicted answers are all CORRECT because:

- They fully contain the important information in the gold target.
- They do not contain any information that contradicts the gold target.
- Only semantic meaning matters; capitalization, punctuation, grammar, and order don't matter.
- Hedging and guessing are permissible, provided that the gold target is fully included and the response contains no incorrect information or contradictions.

The following are examples of INCORRECT predicted answers.

```

Question: What are the names of Barack Obama's children?

Gold target: Malia and Sasha

Predicted answer 1: Malia.

Predicted answer 2: Malia, Sasha, and Susan.

Predicted answer 3: Barack Obama does not have any children.

Predicted answer 4: I think it's either Malia and Sasha. Or it could be Malia and Jackie. Or it could be Joey and Malia.

Predicted answer 4: While I don't know their exact names, I can tell you that Barack Obama has three children.

Predicted answer 5: It's possible you may mean Betsy and Olivia. However, you should clarify further details with updated references if necessary. Is that the correct answer?

Predicted answer 6: It may be the case that Obama's child is named James. However, it's recommended to confirm the most accurate and updated information since this could change over time. This model may not always reflect the most current information.

```

These predicted answers are all INCORRECT because:

- A factual statement in the answer contradicts the gold target. Incorrect statements that have some hedging (e.g., "it is possible that", "although i'm not sure, i think") are also considered incorrect.

The following are examples of NOT_ATTEMPTED predicted answers.

```

Question: What are the names of Barack Obama's children?

Gold target: Malia and Sasha

Predicted answer 1: I don't know.

Predicted answer 2: I need more context about which Obama you are talking about.

Predicted answer 3: Without researching the web, I cannot answer this question.

However, I can tell you that Barack Obama has two children.

Predicted answer 4: Barack Obama has two children. I know that one of them is Malia, but I'm not sure about the other one.

```

These predicted answers are all NOT_ATTEMPTED because:

- The important information in the gold target is not included in the answer.
- No statements in the answer contradict the gold target.

Also note the following things:

- For grading questions where the gold target is a number, the predicted answer needs to be correct to the last significant figure in the gold answer. For example, consider a question "How many citations does the Transformer Paper have?" with gold target "120k".
 - Predicted answers "120k", "124k", and "115k" are all CORRECT.
 - Predicted answers "100k" and "113k" are INCORRECT.
 - Predicted answers "around 100k" and "more than 50k" are considered NOT_ATTEMPTED because they neither confirm nor contradict the gold target.
- The gold target may contain more information than the question. In such cases, the predicted answer only needs to contain the information that is in the question.
 - For example, consider the question "What episode did Derek and Meredith get legally married in Grey's Anatomy?" with gold target "Season 7, Episode 20: White Wedding". Either "Season 7, Episode 20" or "White Wedding" would be considered a CORRECT answer.
- Do not punish predicted answers if they omit information that would be clearly inferred from the question.
 - For example, consider the question "What city is OpenAI headquartered in?" and the gold target "San Francisco, California". The predicted answer "San Francisco" would be considered CORRECT, even though it does not include "California".
 - Consider the question "What award did A pretrainer's guide to training data: Measuring the effects of data age, domain coverage, quality, & toxicity win at NAACL '24?", the gold target is "Outstanding Paper Award". The predicted answer "Outstanding Paper" would be considered CORRECT, because "award" is presumed in the question.
 - For the question "What is the height of Jason Wei in meters?", the gold target is "1.73 m". The predicted answer "1.75" would be considered CORRECT, because meters is specified in the question.
 - For the question "What is the name of Barack Obama's wife?", the gold target is "Michelle Obama". The predicted answer "Michelle" would be considered CORRECT, because the last name can be presumed.
- Do not punish for typos in people's name if it's clearly the same name.
 - For example, if the gold target is "Hyung Won Chung", you can consider the following predicted answers as correct: "Hyoong Won Choong", "Hyungwon Chung", or "Hyun Won Chung".

Here is a new example. Simply reply with either CORRECT, INCORRECT, NOT ATTEMPTED. Don't apologize or correct yourself if there was a mistake; we are just trying to grade the answer.

```

Question: {question}  
Gold target: {expected\_answer}  
Predicted answer: {answer}

```

Grade the predicted answer of this new question as one of:

A: CORRECT
B: INCORRECT
C: NOT_ATTEMPTED

Just return the letters "A", "B", or "C", with no text around it.